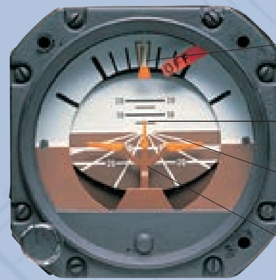
BLIND-FLIGHT TECHNOLOGY

For the first ever blind flight in 1929, test pilot James Doolittle's biplane was equipped with an altimeter 20 times more accurate than the standard devices in use. To replace the turn-and-bank indicator, Elmer Sperry had developed an "artificial horizon." This instrument combined a bar representing the horizon and a small airplane symbol. When the aircraft banked, the horizon bar tilted, and if it changed pitch the bar rose or fell accordingly. Another of Sperry's contributions was a gyrocompass, which, unlike traditional compasses, held stable through turning maneuvers. Doolittle's chief locating device was a radio. He was equipped to receive instructions from a ground controller and to orient himself on a radio beam.



MODERN ARTIFICIAL HORIZON

Originally conceived by Elmer Sperry in 1929, an artificial-horizon instrument shows the angle of an aircraft when banking, as well as its angle of pitch, in degrees.

years later, in 1932, on the fifth anniversary of Lindbergh's crossing – a timing carefully calculated by her publicists – Earhart did the real thing, flying the Atlantic solo in a Lockheed Vega.

The excitement surrounding these flights was above all fueled by risk. Pilots could, and did, still die with appalling frequency, especially when crossing large expanses of ocean. The frisson of death kindled the emotions of the public. But at the same time patient efforts were being made by practical and ingenious inventors and designers precisely to take the risk out of aviation.

Flying blind

One of the most frequent causes of accidents – and the most inhibiting limit on the development of scheduled commercial flights – was the difficulty of flying in low cloud or fog. Poor visibility posed the problem not only of how to find your way, but also of how to keep control of the airplane. Pilots habitually flew by "the seat of their pants," relying on their sense of sight and their instinctive sense of balance. But in fog or dense cloud, with no visual feedback, pilots easily became disoriented. Typically they might misinterpret a banked turn as a dive, pull back on the stick, tighten the turn further, and end up in a fatal spin. In principle, the altimeters and turn-and-bank indicators available since World War I allowed pilots to fly even in zero visibility, but few fliers found it possible to trust these instruments when they contradicted their instincts.

Although the development of improved instruments and instrument-flying was pursued in several countries, notably Germany, some of the most crucial progress was achieved in the late 1920s at the research laboratory established by the Guggenheim Foundation at Mitchel Field, Long Island, New York. Working in conjunction with Elmer Sperry, the world's leading expert on gyroscopic instruments, the Guggenheim Foundation employed racing pilot James Doolittle to experiment with blind flight. The holy grail of their quest was to take off, fly a specific course, and land, without reference to the world outside the cockpit. Doolittle was to attempt the feat in a Consolidated NY-2 modified with a new generation of flying instrumentation (see above).

On September 24, 1929, Doolittle taxied out on to Mitchell Field enclosed by a light-proof hood that cut him off from outside vision. For safety, a second pilot, allowed normal vision, was in the airplane but he did not touch his controls. Doolittle used the radio beam to find the correct line for takeoff, lifted into the air, stayed in the air for a quarter of an hour, making two 180-degree turns, and landed somewhat roughly but safely. The first successful blind flight in history, it was immediately recognized as a major step forward in air safety.

Autopilots

Shortly afterward, Sperry went on to develop the first effective automatic pilot. This was a field in which he had much experience. In 1914, his son Lawrence Sperry had demonstrated a gyrostabilizer at a French airshow. With the pilot's hands off the controls, the device kept a biplane in level, stable flight while a mechanic climbed first onto a wing and then onto the rear fuselage, a shift of weight that should have caused the aircraft to tilt or pitch. No practical use was found for the gyrostabilizer, partly because pilots did not need it and also because it was too unreliable due to the tendency of the gyroscope to "drift" out of alignment. In the 1930s, however, with airmen flying ever-longer distances in airplanes with increasingly complex instruments and radio equipment, the usefulness of a device that could at least temporarily take over control of the airplane became apparent, and Sperry set to work to remedy the defect of drift by linking the gyroscope to a pendulum. In 1933, when one-eyed American pilot Wiley Post made the first solo around-the-world flight, his Lockheed Vega, the *Winnie Mae*, was equipped with a prototype Sperry autopilot.

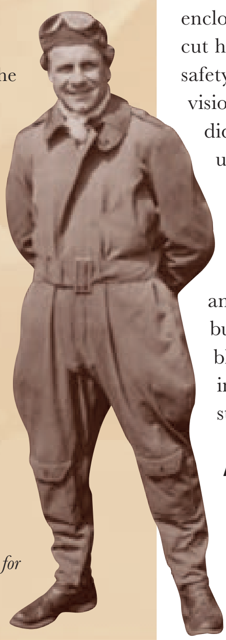
Design breakthroughs

Progress in instrumentation was matched by giant strides in airplane design. The aircraft used by adventurous pilots immediately after World War I – the Vickers Vimy, for example, or the Breguet 14 – were far superior to prewar models in range, engine power, load-carrying capacity, and reliability. But they were still cumbersome strut-and-wire biplanes that looked as if they could have been designed specifically to maximize drag.

The most innovative airframes of the immediate postwar period came from German designers. Both Hugo Junkers and Reinhold Platz, the designer at Fokker, developed aircraft with a single strut-free cantilever wing. Junkers went for all-metal construction, making his monoplanes out of strong, lightweight Duralumin. The metal skin of Junkers aircraft was corrugated to give

JAMES "JIMMY" DOOLITTLE

Born in California, James H. Doolittle (1896–1993) learned to fly with the Army Signal Corps during World War I. In 1922 he became the first man to cross the United States from coast to coast in under 24 hours. Doolittle was one of the most famous showmen of the 1920s, renowned both as a stunt pilot and a record-breaking racer – winner of the Schneider Trophy in 1925, the Bendix Trophy in 1931, and the Thompson Trophy in 1932. But there was also a more serious side to his flying. An exceptional test pilot, he became one of the first people to receive a doctorate in aeronautical engineering in 1925. During World War II, he returned to active service as a senior commander in the USAAF, notably leading a daring long-distance bombing raid on Tokyo in 1942.



PRECISION PILOT

An aviation pioneer who carefully calculated the risks for every stunt, Doolittle lived into his nineties and received almost every major aviation honor.